

A Study of the Time Dependent Ocean Circulation¹

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ABSTRACT

The response of a simple bounded-ocean model to a fluctuating wind stress is studied. Both the linear response and the resulting steady non-linear circulations are computed. The results of this analysis show that the structures of the steady and fluctuating ocean circulations are strongly dependent on the frequency of the forcing.

1. Introduction

The study of the ocean circulation in an enclosed basin has in the past been pivoted around the analysis of the response of a variety of oceanic models to a steady forcing wind stress. (See for example Stommel (1948), Munk (1950), Charney (1955).) In fact, the wind forcing, especially in mid-latitudes, is notoriously unsteady. Aside from an interesting numerical study by Veronis (1963) very little attention has been directed toward the unsteady problem. In this paper *only* the unsteady component of the atmospheric stress is considered.

The importance of such a study is that in addition to the direct fluctuating response of the ocean, an oscillatory wind forcing, with a *zero* time mean, will produce a steady time independent circulation due to the non-linearity of the governing dynamics. The implication is that the total *steady* oceanic circulation cannot be described by only the steady component of the wind forcing.

This paper's purpose is to explore some of the possible fluctuating and steady circulations produced in a simple model by time dependent forcing. Particular attention will be paid to the dependence of the resulting circulations on the characteristics of the wind forcing and the ocean's internal parameters.

2. The model

In order to focus attention on the complicating factor of the fluctuating character of the forcing, the ocean model used is quite simple; it is a uniform and homogeneous sheet of fluid on the β plane, on which the fluid motion is two-dimensional and incompressible. The forcing field is considered as a body force, while the viscous dissipation is modeled by a simple velocity drag law. With these assumptions the governing equation for the stream function is the statement of local vorticity

balance

$$\nabla^2 \psi_t + \beta \psi_x + \psi_x \nabla^2 \psi_y - \psi_y \nabla^2 \psi_x = \frac{\nabla x \tau}{\rho_0 H} - R \nabla^2 \psi. \quad (2.1)$$

τ is the forcing wind stress, ρ_0 the density of the fluid, H its constant depth while R is the coefficient of viscous drag. β is the northward gradient of the coriolis parameter. The co-ordinate system is arranged so that x measures positive eastward, y is measured positive northward. The eastward velocity u and the northward velocity v are derived from ψ by the relations

$$u = -\psi_y \\ v = \psi_x.$$

For simplicity, although not necessary for the analysis, the ocean is assumed to be square of side length L , on the boundaries of which a zero normal flow condition is imposed, i.e.,

$$\psi = 0 \text{ on } x=0, \quad x=L, \quad y=0, \quad y=L. \quad (2.2)$$

It is convenient to introduce non-dimensional variables in the following manner. If τ_0 is the characteristic amplitude of the wind stress, the non-dimensional variables are defined by the relations:

$$\begin{aligned} \nabla x \tau &= \frac{\tau_0}{\rho_0 H L} T(x', y', t') \\ (x, y) &= L(x', y') \\ t &= (\beta L)^{-1} t' \\ \psi &= \frac{\tau_0}{\rho_0 \beta H} \Phi. \end{aligned} \quad (2.3)$$

The vorticity equation may then be rewritten in terms of these variables after dropping the prime notation for the dimensionless independent variables.

$$\nabla^2 \Phi_t + \Phi_x + \text{Ro}(\Phi_x \nabla^2 \Phi_y - \Phi_y \nabla^2 \Phi_x) = T(x, y, t) - \delta \nabla^2 \Phi. \quad (2.4)$$

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Two non-dimensional parameters appear; the Rossby number $Ro \equiv \tau_0 / \rho_0 \beta^2 H L^3$ and a dimensionless viscosity $\delta = R / \beta L$. The central assumption made to simplify the analysis is the weakening assumption that Ro is very small, and indeed that $Ro\delta^{-1}$ is of the same order as Ro . If this assumption is made, the lowest order dynamics are purely linear and the non-linear effects can be computed by a perturbation series in Ro . Certainly this is a restrictive assumption but one whose real justification is by the explicitness and accessibility of the solution. Φ is thus expanded in the series:

$$\Phi = \sum_{n=0} Ro^n \Phi^{(n)}. \quad (2.5)$$

The lowest order problem, for $\Phi^{(0)}$, is the linear problem:

$$\nabla^2 \Phi_t^{(0)} + \Phi_x^{(0)} + \delta \nabla^2 \Phi^{(0)} = T(x, y, t). \quad (2.6)$$

The first non-linear correction is described by the equation for $\Phi^{(1)}$

$$\nabla^2 \Phi_t^{(1)} + \Phi_x^{(1)} + \delta \nabla^2 \Phi^{(1)} = -(\Phi_x^{(0)} \nabla^2 \Phi_y^{(0)} - \Phi_y^{(0)} \nabla^2 \Phi_x^{(0)}). \quad (2.7)$$

In the first order problem the non-linear vorticity advection of the zero order fields plays a similar role as the wind stress does for the zero order problem. In this paper we shall settle for computing $\Phi^{(0)}$ and $\Phi^{(1)}$, the linear response and the first non-linear correction.

3. The linear problem

To satisfy the zero boundary conditions at $y=0$ and $y=1$ the stream function $\Phi^{(0)}$ is expanded, along with the forcing field T , as a sine series

$$a) \quad \Phi^{(0)} = \sum_{n=1}^{\infty} \varphi_n \sin n\pi y, \quad (3.1)$$

$$b) \quad T = \sum_{n=1}^{\infty} \tau_n \sin n\pi y.$$

The governing differential equation for φ_n is:

$$\frac{\partial}{\partial t} (\varphi_{nxx} - n^2 \pi^2 \varphi_n) + \varphi_{nx} + \delta (\varphi_{nxx} - n^2 \pi^2 \varphi_n) = \tau_n. \quad (3.2)$$

With the use of the Laplace transform apparatus it can be easily shown that the general solution for φ_n can

be written

$$\varphi_n = \frac{1}{2\pi i} \int_{-\infty-i}^{\infty-i} dp \int_0^1 dx' \exp\left(\frac{x'-x}{2(p+\delta)} + pt\right) \times \frac{\sinh \alpha_n x_{<} \sinh \alpha_n (x_{>} - 1)}{(p+\delta) \alpha_n \sinh \alpha_n} [\hat{\tau}_n + V_n]. \quad (3.3)$$

V_n is the initial value of the n^{th} component of the vorticity, $\hat{\tau}_n$ is the Laplace transform of τ_n , while

$$\alpha_n = \frac{1}{2(p+\delta)} \sqrt{1 + 4n^2 \pi^2 (p+\delta)^2} \quad (3.4)$$

p of course is the dummy Laplace variable, $x_{<}$ is the smaller of x and x' while $x_{>}$ is the larger.

4. The normal modes

Before discussing forced motions it is useful to consider the free ($\tau_n = 0$) oscillations in the bounded ocean. For $\tau_n = 0$ the only solutions are those produced by the initial conditions. To evaluate (3.3) we note that the integrand has a pole wherever $\alpha_n^2 = -m^2 \pi^2$ ($m = 1, 2, \dots$). There is no pole at $p = -\delta$. The only poles are at

$$p = p_{mn} = -\delta \pm \frac{i}{2\pi} (\sqrt{m^2 + n^2})^{-1} \quad m = 1, 2, \dots \quad (4.1)$$

The time dependence of the normal modes consists of a decay time δ^{-1} (independent of scale) and an oscillation frequency which diminishes as the labeling indices m and n (roughly corresponding to the inverse scale of the disturbance) become large. This curious frequency law arises because the smallest scales of motion do not "see" the variation of coriolis parameter and therefore do not oscillate. The normal mode poles lie on the line $p = -\delta$ in the Laplace transform plane and for each n lie between $p = \pm i/2\pi \sqrt{n^2 + 1}$. There is a point of accumulation of the poles at $p = -\delta + i0$ in which neighborhood there are an infinite number of poles. It is not difficult to show however, that (3.3) may still be evaluated by taking the residue at each pole in the formal manner, as though the poles were well separated. The proof is a straightforward examination of the integrand of (3.3) near $p = -\delta$ and is omitted here. The residue calculation then yields for (3.3)

$$\varphi_n = 4 \sum_{m=1}^{\infty} e^{-\delta t} \cos\left(\frac{x}{2\omega_{mn}} + \omega_{mn} t\right) \sin m\pi x \sin n\pi y \int_0^1 dx' V_n(x') \sin m\pi x' \cos \frac{x'}{2\omega_{mn}} \\ + 4 \sum_{m=1}^{\infty} e^{-\delta t} \cos\left(\frac{x}{2\omega_{mn}} + \omega_{mn} t\right) \sin m\pi x \sin n\pi y \int_0^1 dx' V_n(x') \sin m\pi x' \sin \frac{x'}{2\omega_{mn}}, \quad (4.2)$$

where

$$\omega_{mn} = \frac{1}{2\pi\sqrt{m^2+n^2}}.$$

By inspection the normal modes which remove the original disturbance have the form:

$$\varphi_n^m = e^{-\delta t} \left\{ \begin{array}{l} \cos\left(\frac{x}{2\omega_{mn}} + \omega_{mn}t\right) \\ \sin\left(\frac{x}{2\omega_{mn}} + \omega_{mn}t\right) \end{array} \right\} \sin m\pi x \sin n\pi y. \quad (4.3)$$

Each normal mode consists of a traveling wave (with phase speed always toward the west) modulated by an envelope of sine functions which satisfy the boundary conditions. Each mode has both fixed and moving nodes and the circulations in each mode will show cells of vortical motion, each cell moving westward, alternately increasing and decreasing in size as a movable node approaches and then passes a fixed node. Within a time

$t = \delta^{-1}$ the initial conditions will disappear. It is interesting to note that aside from a preferred direction for the phase speed there is no east-west asymmetry in the solutions. This structure of the normal modes will reappear in the forced problem which is studied below.

5. Forced solutions

We will now study the effect of one forcing function which will consist of a traveling wave passing over the ocean whose form is

$$T = \cos(kx - \omega t) \sin n\pi y. \quad (5.1)$$

This stress field has three disposable parameters, k , ω and n , and the reaction of the resultant ocean circulation as a function of these parameters, particularly ω is our primary concern. Using (3.3) the solution for $\varphi_n^{(0)}$ can be shown to consist of a linear combination of normal modes necessary to satisfy initial conditions plus the appropriate particular forced solution. After the normal modes decay only the forced solution remains and can be written

$$\varphi_n = -\text{Re} \left[\exp i(kx - \omega t) + \frac{\sinh \alpha_n (x-1)}{\sinh \alpha_n} \exp \left(-i\omega t - \frac{ix}{2(\omega + i\delta)} \right) - \frac{\sinh \alpha_n x}{\sinh \alpha_n} \exp \left(-i\omega t + \frac{(1-x)i}{2(\omega + i\delta)} + ik \right) \right] \frac{1}{\kappa_n^2 (\delta - i\omega)}. \quad (5.2)$$

In this formula

$$\alpha_n = \frac{1}{2(\delta - i\omega)} \sqrt{1 + 4n^2\pi^2(\delta - i\omega)^2},$$

$\kappa_n^2 = (k^2 + n^2\pi^2) + k/(\omega + i\delta)$ while Re refers to the real part of the right hand side of (5.2). Although a discussion of the solution (5.2) can be given in general, the form of the solution is sufficiently variable so that it is more illuminating, especially in the discussion of the non-linear circulations, to consider certain special cases.

Case 1 Resonance ($\omega = \omega_{mn}$)

Perhaps one of the most important cases occurs when the frequency of the forcing function lies within the range of the characteristic frequencies of the ocean basin. Indeed when $\omega = \omega_{mn}$ the system resonates and the amplitude of the linear response is limited only because of the presence of dissipation. If δ is small the linear response, to the lowest order in δ , can be written

$$\Phi^{(0)} = \frac{m \sin m\pi x}{\pi(m^2 + n^2)\kappa_n^2\delta} \left[\cos \left(\omega_{mn}t + \frac{x}{2\omega_{mn}} \right) - (-1)^m \cos \left(\omega_{mn}t + \frac{x-1}{2\omega_{mn}} - k \right) \right] \sin n\pi y. \quad (5.3)$$

The linear response at resonance is a combination of two normal modes, the phase lag between them depending on k and ω_{mn} . The resonant response decreases in amplitude as the frequency of the driving force decreases. The amplitude of the linear solution also decreases with the scale of the disturbance so that in general the larger the scale and the higher the frequency of the disturbing wind field (as long as we are at one of the resonant frequencies!) the greater the response. The configuration of the solution is the same as for the normal modes. We note again that except for a preferred direction for the phase speed there is no asymmetry in the solution between east and west, unlike the steady linear solution of Stommel (1948).

To derive the first non-linear effects we must refer to equation (2.8). The Jacobian of the zero order stream function and vorticity must be computed and the particular solution of (2.8) for that inhomogeneity must be found. Since the Jacobian is a quadratic function of the amplitudes there will be in general an inhomogeneity which oscillates at frequency $2\omega_{mn}$ and a steady *time independent* component. It is the latter that is of interest for that steady component of the Jacobian will force a time independent circulation in the first order problem. If the subscript s denotes the time independent part of any function:

$$\delta \nabla^2 \Phi_s^{(1)} + \Phi_{sx}^{(1)}$$

$$= -(\Phi_x^{(0)} \nabla^2 \Phi_y^{(0)} - \Phi_y^{(0)} \nabla^2 \Phi_x^{(0)})_s. \quad (5.4)$$

The evaluation of the steady part of the Jacobian of the solution (5.3) yields

$$\delta \nabla^2 \Phi_s^{(1)} + \Phi_{sx}^{(1)} = -\frac{\pi^2 n m^3 \sin 2m\pi x \sin 2n\pi y}{(m^2 + n^2) \kappa_n^4 \delta^2} \left\{ 1 - (-1)^m \cos \left(k + \frac{1}{2\omega_{mn}} \right) \right\}. \quad (5.5)$$

We observe the following interesting fact. Since the x integral of the right hand side of (5.5) from $x=0$ to $x=1$ is identically zero, the solution of (5.5), to within an error of order δ (considered small here), can be found by balancing the second term on the left hand side against the non-linear forcing, in complete analogy with the so called Sverdrup open ocean solution for the steady problem. Only in this case, the Sverdrup solution also satisfies the boundary conditions. Thus to this order in δ ,

$$\Phi_s^{(1)} = -\frac{nm^2\pi}{(m^2 + n^2) \kappa_n^4 \delta^2} \sin^2 m\pi x \sin 2n\pi y \left\{ 1 - (-1)^m \cos \left(\frac{1}{2\omega_{mn}} + k \right) \right\}. \quad (5.6)$$

The nature of this steady circulation is interesting. First of all there is no boundary layer character, or east-west asymmetry in this steady circulation, in distinction to the steady circulation produced by a steady forcing. Consider the first mode, $m=n=1$. This consists of two cells, one north of $y=\frac{1}{2}$ the other south of $y=\frac{1}{2}$. The circulation in the southern cell is counterclockwise, that in the northern clockwise. The senses of these circulations are unambiguous and depend only on the structure of the normal modes and not on the parameters of the forcing. The steady circulation produced at resonance always provides, for the first mode, a zonal flow toward the west at $y=\frac{1}{2}$. The direction of the zonal flow at $y=\frac{1}{2}$ then alternates as n increases and more cells are added in the y direction. It is also interesting to note that $\Phi_s^{(1)}$ not only satisfies the condition that the normal flow be zero at $x=0$ and 1 but that the tangential velocity is also zero at both eastern and western boundaries, in sharp distinction with the ordinary boundary layer character of the purely steady problem.

We also note that the steady circulation *vanishes* for certain special values of k . When

$$k = k_j = \pi(\sqrt{m^2 + n^2} \pm j) \quad j \neq 0 \quad (5.7)$$

such that $m+j$ is even, the steady circulation disappears.

Case 2 $\omega \neq \omega_{mn}, \omega \gg \delta$

When the forcing frequency is much larger than δ , and not a resonant frequency the linear solution (5.2) to lowest order in δ/ω is:

$$\Phi^{(0)} = \frac{1}{\omega \kappa_n^2} \left[\sin(kx - \omega t) - \sin \left(\omega t + \frac{x}{2\omega} \right) \frac{\sinh \alpha_n (x-1)}{\sinh \alpha_n} + \sin \left(\omega t + \frac{x-1}{2\omega} - k \right) \frac{\sinh \alpha_n x}{\sinh \alpha_n} \right] \sin n\pi y. \quad (5.8)$$

Thus when the forcing is non-resonant the linear response consists of three parts. The first part, given by the first term on the right hand side of (5.8) mirrors the traveling forcing wave and moves to the east. The other two parts closely resemble the form of the normal modes. The phase of each of these two contributions moves westward opposite to the phase direction of the forcing. Each of the last two effects, which are produced as reactions from the boundary, are modulated by a function whose structural character is determined by the frequency of the forcing function. If $\omega < (2n\pi)^{-1}$ then α_n is imaginary, the modulating functions are oscillatory and the boundary reactions to the forcing are felt everywhere in the ocean interior. On the other hand at high frequencies such that $\omega > (2n\pi)^{-1}$, α_n is real and the reactions from the boundaries, especially for large n decay exponentially away from the boundaries. The cross-over frequency $\omega = (2n\pi)^{-1}$ is the maximum frequency for a free Rossby wave whose y wave number is fixed at $(n\pi)$. Thus if the ocean is forced by an oscillating disturbance whose frequency is so high that no Rossby wave can locally propagate at that frequency the solution exhibits a boundary layer structure for the envelope of the westward propagating component. Note that just as in the case of resonance there is no asymmetry in the solution between east and west. *At high frequencies boundary layers occur at both eastern and western walls.*

The non-linear steady circulation in the non-resonant case shares the property with the resonant case that the x integral of the Jacobian of the zero order stream function and vorticity from $x=0$ to $x=1$ vanishes. The Sverdrup solution thus satisfies the boundary conditions at both $x=0$ and $x=1$, so that no asymmetry is to be expected in the steady circulation.

The steady circulation in this case is parametrically quite complicated and only the solution for the purely zonal forcing ($k=0$) will be displayed. The steady circulation is:

$$\begin{aligned} \Phi_s^{(1)} = & -\frac{n\pi}{4(\omega\kappa_n^2)^2} \sin 2n\pi y \left[\frac{1}{\kappa_n^2} \left\{ \sin \frac{x}{2\omega} \cosh \alpha_n(x-1) - \sin \frac{(x-1)}{2\omega} \cosh \alpha_n x - \sin \frac{1}{\omega} \right\} \frac{n^2\pi^2\alpha_n}{\omega} \right. \\ & + \frac{1}{\kappa_n^2} \left\{ \sinh \alpha_n x \cos \frac{(x-1)}{2\omega} - \sinh \alpha_n(x-1) \cos \frac{x}{2\omega} - \sinh \alpha_n \right\} n^2\pi^2 \left(\frac{n^2\pi^2}{2\omega^2} - \frac{1}{2\omega^2} \right) \\ & \left. + \frac{1}{4\omega^2 \sinh^2 \alpha_n} \left\{ (\cosh 2\alpha_n(x-1) - \cosh \alpha_n) + (\cosh(2\alpha_n x) - 1) - 2 \cos \frac{1}{2\omega} (\cosh \alpha_n(2x-1) - \cosh \alpha_n) \right\} \right]. \quad (5.9) \end{aligned}$$

The steady circulation, like the linear oscillating solution changes character where $\omega = (2n)^{-1}$. At high frequencies $\omega \gg (2n)^{-1}$ an approximate form of (5.9) may be written:

$$\Phi_s^{(1)} = \frac{\sin 2n\pi y}{4\omega^2 n^2 \pi^2} [1 - e^{n\pi(x-1)} - e^{-n\pi x}].$$

Thus at high frequencies there exist two steady boundary layers, *one on each side of the ocean*. For $n=1$ the solution consists again of two cells, one in the upper half of the ocean and one in the lower half. The sense of the circulation in each cell is reversed from the resonant case. The cell in the lower half now rotates clockwise while the one in the upper half rotates counterclockwise. The flow in each cell is almost purely zonal away from the boundaries while near the boundaries the flow is intense and meridional. As n increases the number of cells increase, the boundary layer width decreases, and the amplitude of the steady circulation decreases. The circulation also decreases in magnitude as ω^2 increases in distinction to the resonant case. Again the *sense* of the circulation for each n is given unambiguously.

In the case where $\omega < (2n\pi)^{-1}$, α_n is imaginary. The circulation does not have a boundary layer character but is highly oscillatory, especially for fairly small ω . In general the sense of the steady circulation at any point will oscillate with ω and in this parameter region the sense of the circulation, unlike the above cases is ambiguous.

Case 3 $\omega \ll (2n\pi)^{-1}$; $\omega = 0(\delta)$

In this last case we will look at low frequency forcing for which the time scale of the forcing is much longer than the characteristic time scale of the normal modes and is of the order of the viscous time scale δ^{-1} . In this case the appropriate form for the linear solution (5.2) is (with the notation $s^2 = 2(\delta^2 + \omega^2)$)

$$\begin{aligned} \Phi^{(0)} = & \frac{\sin n\pi y}{k} \left\{ \sin(kx - \omega t) + \sin\left(\omega t + \frac{2\omega x}{s^2}\right) e^{-(2\delta x/s^2)} \right. \\ & \left. + \sin(\omega t - k) - \sin\left(\omega t - k + \frac{2\omega x}{s^2}\right) e^{-(2\delta x/s^2)} \right\}. \quad (5.10) \end{aligned}$$

This solution has a very interesting structure. The solution consists of two main parts. The first part is a traveling wave moving in the direction of the forcing wave and similar in form to the forcing while the rest of the solution is an oscillating boundary layer, this time with the boundary layer *only on the western side of the ocean*. This oscillating boundary layer consists of the by now familiar westward propagating phase modulated by an exponential decay. If we let k and ω both vanish we recover the linear boundary layer solution for steady zonal forcing first found by Stommel (1948).

$$\Phi^{(0)} = [(x-1) + e^{-x/\delta}] \sin n\pi y. \quad (5.11)$$

If we now compare the boundary layer structure of the oscillating solution to that of the steady solution we may note the following. The characteristic scale of the oscillating boundary layer, x_0 , is *larger* than the scale of the steady boundary layer x_s . The ratio of the scales being

$$\frac{x_0}{x_s} = 1 + \frac{\omega^2}{\delta^2}. \quad (5.12)$$

Let us now consider the first order non-linear problem. Again for algebraic simplicity we restrict ourselves to zonal forcing ($k=0$). We obtain:

$$\begin{aligned} \delta^2 \Phi_s^{(1)} + \Phi_{sx}^{(1)} = & \left\{ e^{-(4\delta x/s^2)} \left[\frac{32\delta\omega^2}{s^6} \right] \right. \\ & - \cos \frac{2\omega x}{s^2} e^{-(2\delta x/s^2)} \left[(1-x) \frac{8\delta(\omega^2 - \delta^2) + 16\omega^2\delta}{s^6} \right] \\ & \left. - \sin \frac{2\omega x}{s^2} e^{-(2\delta x/s^2)} \left[(1-x) \frac{8\omega(\omega^2 - \delta^2) + 16\omega\delta^2}{s^6} \right] \right\} \\ & \times (\sin 2n\pi y) \frac{n\pi}{4}. \quad (5.13) \end{aligned}$$

The integral of this forcing term from $x=0$ to $x=1$ is approximately $n\pi/2s^2 \sin 2n\pi y$. Thus the Sverdrup solution for this forcing will *not* satisfy the boundary conditions on the western edge of the ocean. In order to find the steady flow it is necessary to solve (5.13) completely. While not a difficult task the algebra is sufficiently complicated to obscure the result so that a special case will

be chosen in which $\delta = \omega$ exactly. In this case (5.13) becomes approximately

$$\delta' \Phi_{xx}^{(1)} + \Phi_x^{(1)} = \left[-\frac{1}{2} \frac{e^{-x/\delta}}{\delta^3} + \frac{1}{4\delta^3} e^{-x/2\delta} \right] \times \left\{ \sin \frac{x}{2\delta} - \cos \frac{x}{2\delta} \right\} \frac{n\pi}{4} \sin 2n\pi y \quad (5.14)$$

which has as a solution

$$\Phi^{(1)} = \left[\left(\cos \frac{x}{2\delta} - \sin \frac{x}{2\delta} \right) e^{-x/2\delta} - \left(\frac{x}{\delta} + 1 \right) e^{-x/\delta} \right] \frac{n\pi \sin 2n\pi y}{4 \cdot 2\delta^2}. \quad (5.15)$$

This steady flow consists of two parts. The first part has a boundary layer thickness twice the linear steady one; at the same time this fatter boundary layer spatially oscillates very rapidly. The second part of the solution is non-oscillatory and has the same scale as the linear steady solution of Stommel (1948). The boundary layer structure is more readily revealed by considering the northward velocity $\Phi_x^{(1)}$

$$\Phi_x^{(1)} = -\frac{1}{\delta^3} \left[\frac{x}{\delta} e^{-x/\delta} - \cos \frac{x}{2\delta} e^{-x/2\delta} \right] \frac{n\pi}{4} \sin 2n\pi y. \quad (5.16)$$

For $n=1$ the tangential velocity at the western boundary in the lower half of the ocean is negative and

then exponentially decreases in amplitude, but oscillates in sign, oceanward. There exists therefore a cell of counterclockwise rotation in the bottom half of the ocean, clockwise rotation in the upper half, squeezed in a very narrow region near the western boundary. Although this computation has been done for only one value of the ratio ω/δ the fact that the integral of the right hand side of (5.13) is a positive constant $n\pi/2s^2$ multiplied by $\sin 2n\pi y$ means that the *sense* of the circulation is preserved for all values of ω/δ although the details may change. Thus a time dependent forcing of low frequency will produce a narrow boundary current region, which has zero net northward transport because of its highly oscillatory character.

The interesting variation in the structure of the non-linear steady circulations with forcing frequency strongly suggests that a model of the oceanic circulation utilizing only the mean winds is by no means an adequate one. The complete power spectrum of the wind forcing is needed to describe the steady oceanic circulation, since questions concerning the symmetry, structure, sense, and strength of the circulations are strongly frequency dependent.

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